

2078

Bachelor of Science in Computer Science and Information Technology

Course Title: **Numerical Method**

Code No: **CSC212** Semester: **3rd Semester**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. How can Horner's rule be used to evaluate the $f(x)$ and $f'(x)$ of a polynomial at given point? Explain. Write an algorithm and program to calculate a real root of a polynomial using Horner's rule.
2. What is matrix factorization? How can it be used to solve system of linear equations? Factorize the given matrix A and solve the system of equations $Ax = b$ for given b using L and U matrices. $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 1 & 1 \\ 2 & 3 & 2 \end{bmatrix}$ and $b = \begin{bmatrix} 2 \\ 8 \\ 28 \end{bmatrix}$
3. What is higher order differential equation? How can you solve the higher order differential equation? Explain. Solve the following differential equation for $1 \leq x \leq 2$, taking $h = 0.25$. $\frac{d^2y}{dx^2} + 3 \frac{dy}{dx} + 5y = 0$, with $y(1) = 1$ and $y'(1) = 2$.

Section B

Attempt any Eight questions [8×5=40]

4. How the half-interval method can estimate a root of non-linear equation? Find a real root of following equation using half-interval method correct up to two decimal places. $x^2 - e^x - x = 1$
5. Calculate a real root of the given equation using fixed point iteration correct up to 3 significant figures. $2x^3 - 2x = 5$
6. What is Newton's interpolation? Obtain the divided difference table from the following data set and estimate the $f(x)$ at $x = 2$ and $x = 5$.

x	1.0	1.21	1.41	1.61	1.82	2.02	2.22	2.4
$f(x)$	2.02	0.63	0.96	0.09	0.31	5.20	6.30	4.4
7. What is linear regression? Fit the linear function to the following data.

x	1.0	1.21	1.41	1.61	1.82	2.02	2.22	2.4
y	2.02	0.63	0.96	0.09	0.31	5.20	6.30	4.4
8. What are the problems with polynomial interpolation for large number of data set? How such problems are addressed? Explain with example.
9. Evaluate the following integration using Romberg integration. $\int_0^1 (\sin^2 x)/x \, dx$
10. Solve the following set of linear equations using Gauss Jordan method.
$$\begin{aligned} x + 2x + 3x &= 97x + 6x + 5x \\ + 4x &= 338x + 9x + x = 272x + 5x + 4x + 3x = 23 \end{aligned}$$

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- 11.** Solve the following differential equation for $1 \leq x \leq 2$, taking $h = 0.25$ using Heun's method. $y'(x) + x^2y = 3x$, with $y(1) = 1$
- 12.** Consider a metallic plate of size 90 cm by 90 cm. The two adjacent sides of the plate are maintained at temperature of 100°C and remaining two adjacent sides are held at 200°C . Calculate the steady state temperature at interior points assuming a grid size of 30 cm by 30 cm.